

CIS 371 Web Application Programming

Docker



Lecturer: **Dr. Haoyu Li**



What is Docker?

- ❖ **What is Docker**
- ❖ **Virtual Machines vs Containers**
- ❖ **Docker Architecture and Workflow**
- ❖ **Installing Docker**
- ❖ **Development Workflow**

What is Docker?

A platform for **building**, **running**, and **shipping** applications.

What is Docker?

Situation: Your application works on your development machine but doesn't somewhere else. Why?



What is Docker?

Situation: Your application works on your development machine but doesn't somewhere else. Why?

- **One or more files missing**

What is Docker?

Situation: Your application works on your development machine but doesn't somewhere else. Why?

- **One or more files missing**
- **Software version mismatch**

What is Docker?

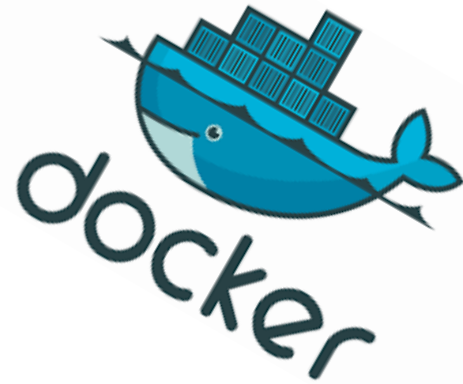
Situation: Your application works on your development machine but doesn't somewhere else. Why?

- **One or more files missing**
- **Software version mismatch**
- **Different configuration settings**

What is Docker?

Situation: Your application works on your development machine but doesn't somewhere else. Why?

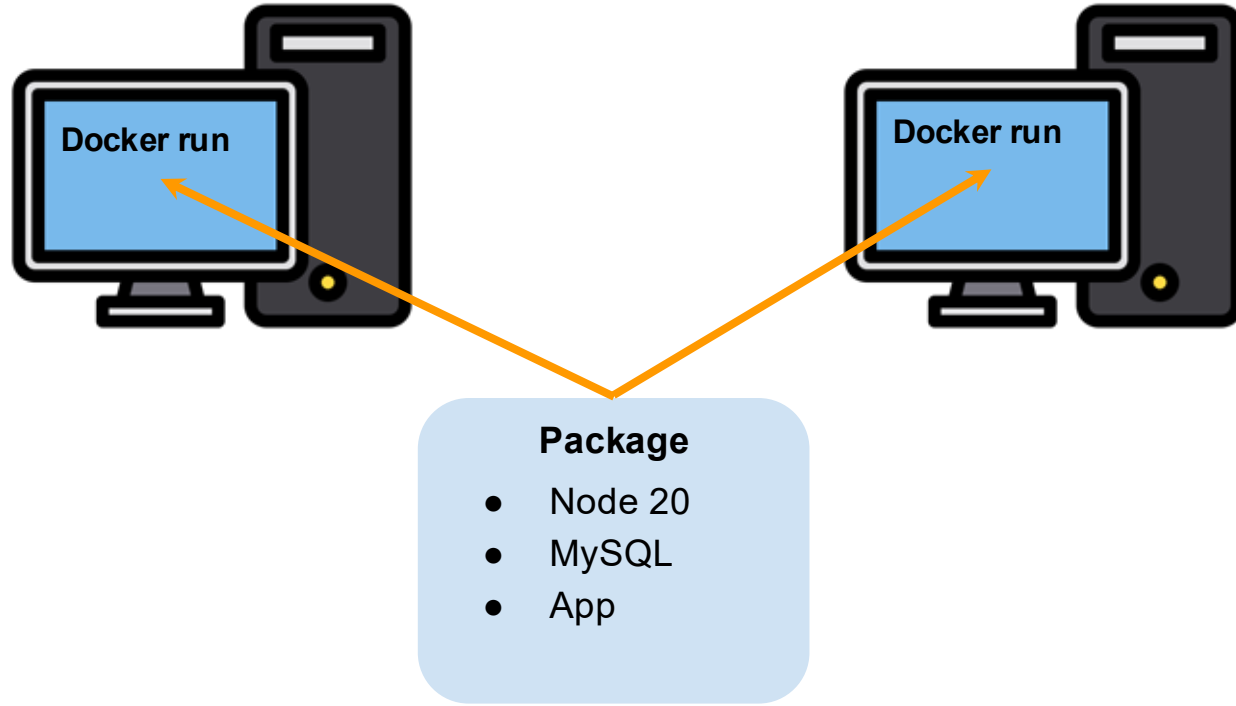
- One or more files missing
- Software version mismatch
- Different configuration settings



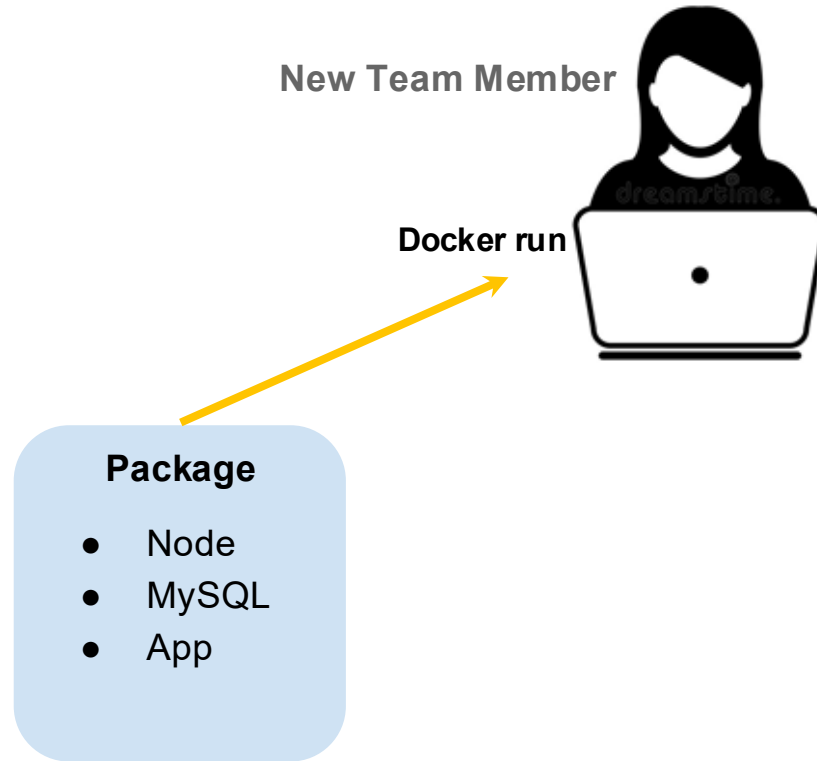
Advantage: Portability & Faster Deployments

Development Machine

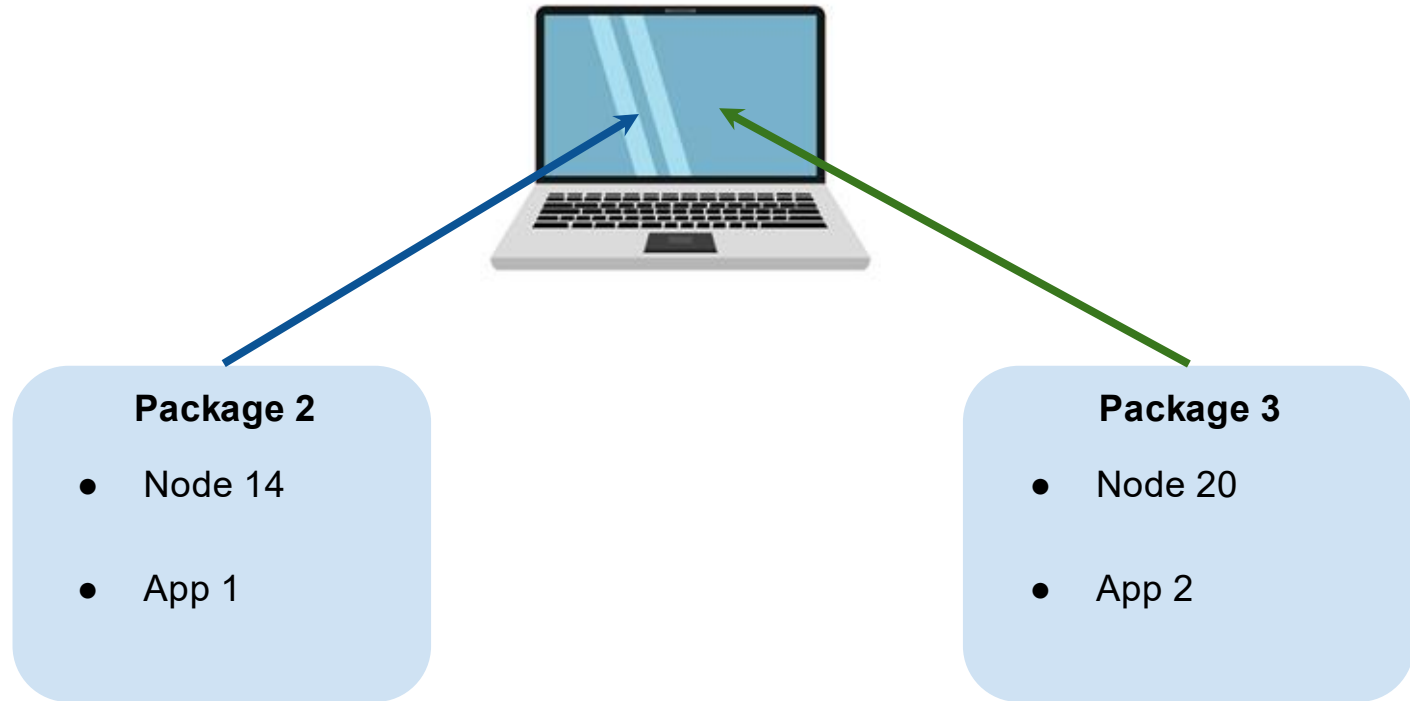
Test or Production Machine



Advantage: Sharing & Simplified Dependency



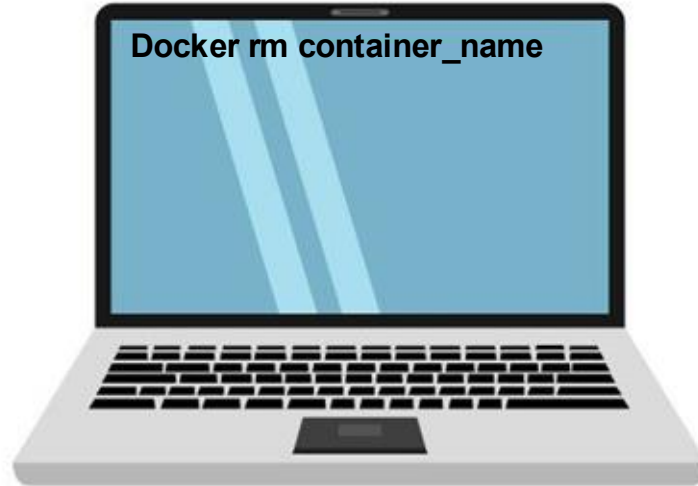
Advantage: Isolation & Security Sandbox



One more benefit



What is Docker?



What is Docker?

A platform for
Consistently
building, running, and shipping applications.



Build

Develop an app using Docker containers with any language and any toolchain.



Run

Scale to 1000s of nodes, move between data centers and clouds, update with zero downtime and more.



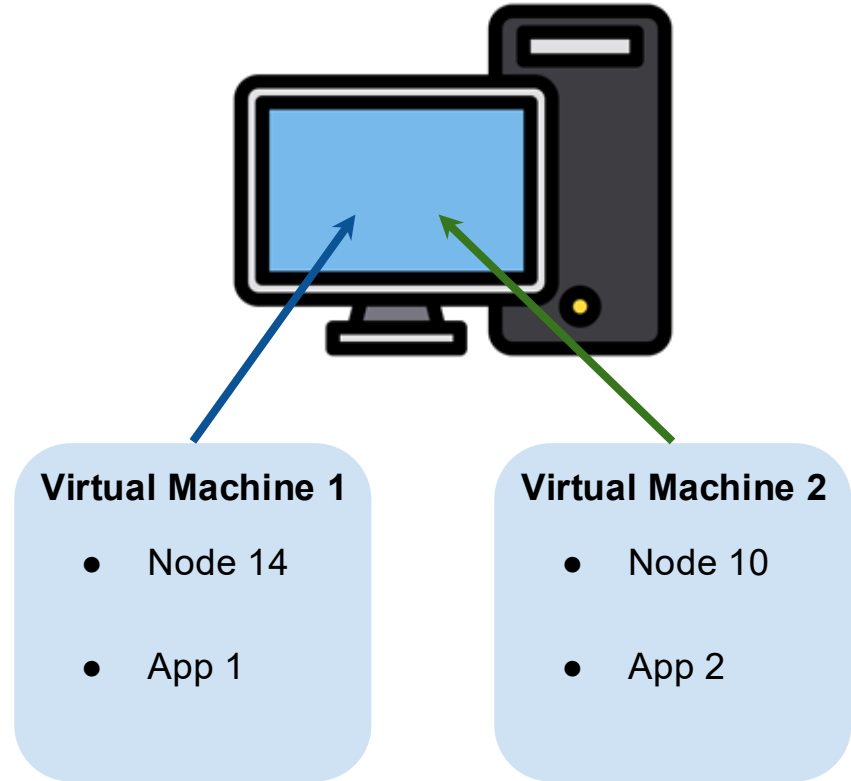
Ship

Ship the "Dockerized" app and dependencies anywhere - to QA, teammates, or the cloud - without breaking anything.

Virtual Machines

Virtual Machine

An abstraction of a machine
(physical hardware)



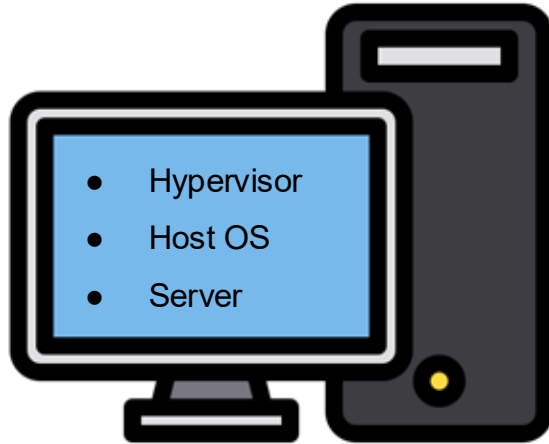
Virtual Machines

VM 1

- App 1
- Bins/Libs
- Guest OS

VM 2

- App 2
- Bins/Libs
- Guest OS



Problems

- Each VM needs a OS
- Slow to start
- Resource intensive

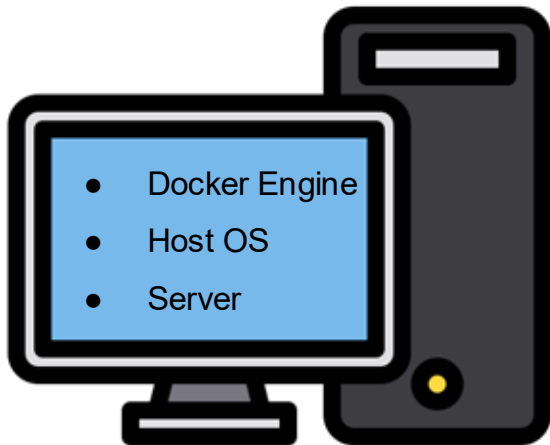
Docker Containers

Container 1

- App 1
- Bins/Libs

Container 2

- App 2
- Bins/Libs

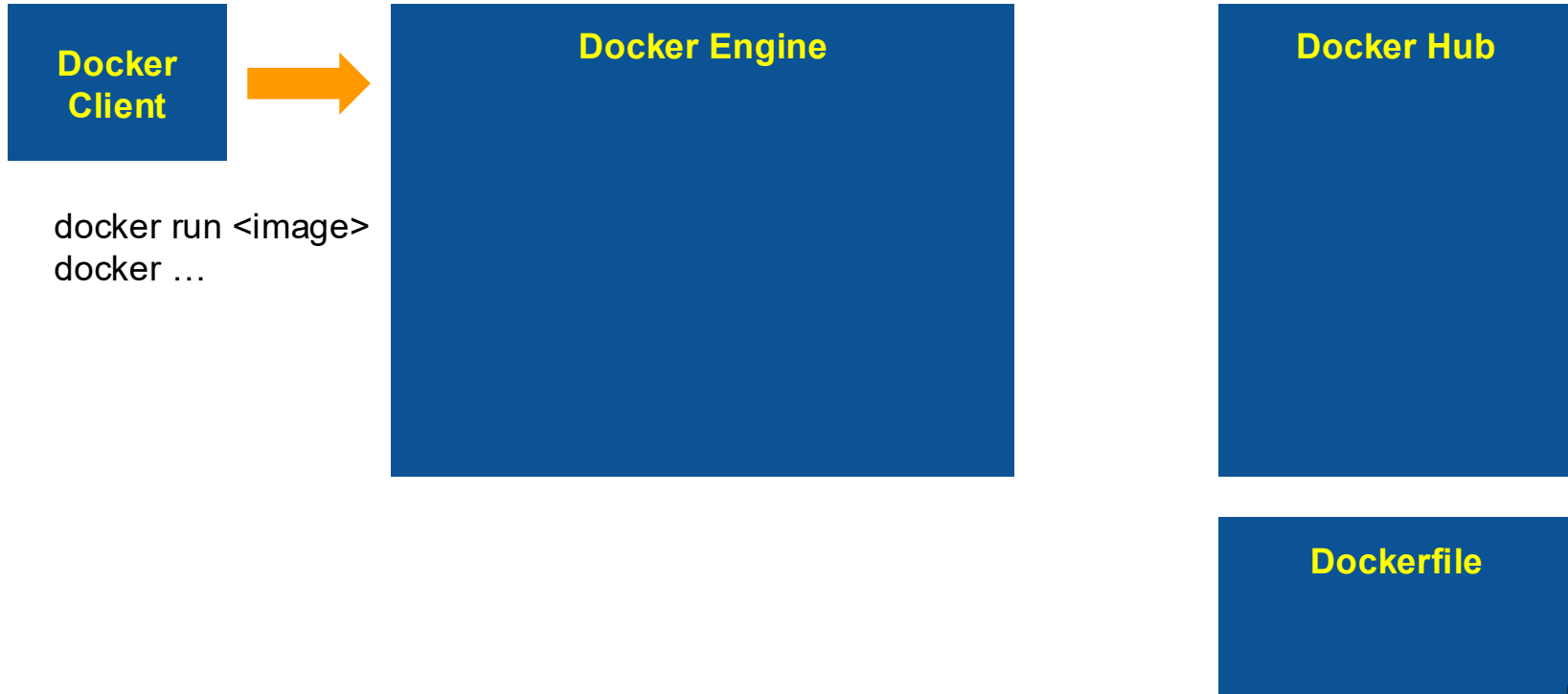


Docker Container

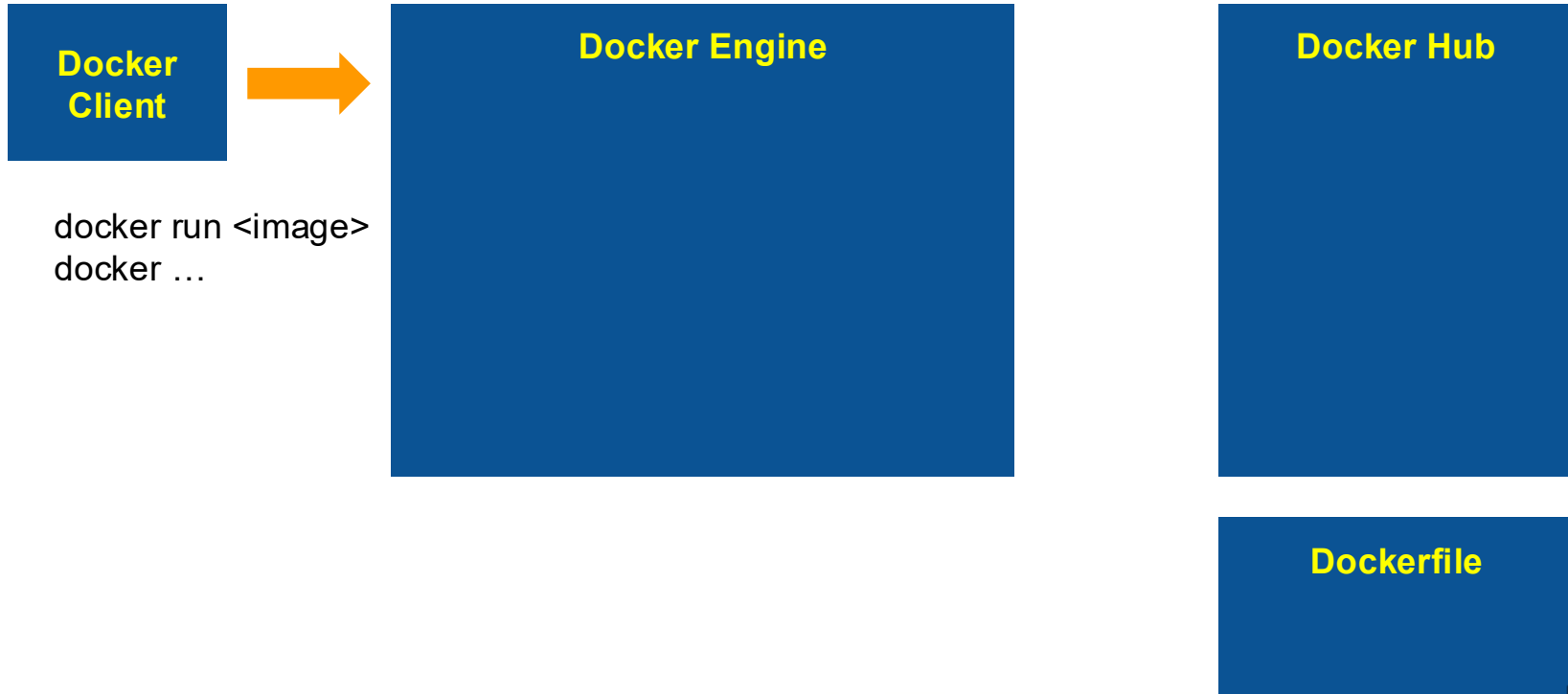
An isolated environment for running an application

- **Allow running multiple apps in isolation**
- **Are lightweight**
- **Use OS of the host**
- **Start quickly**
- **Need less hardware resources**

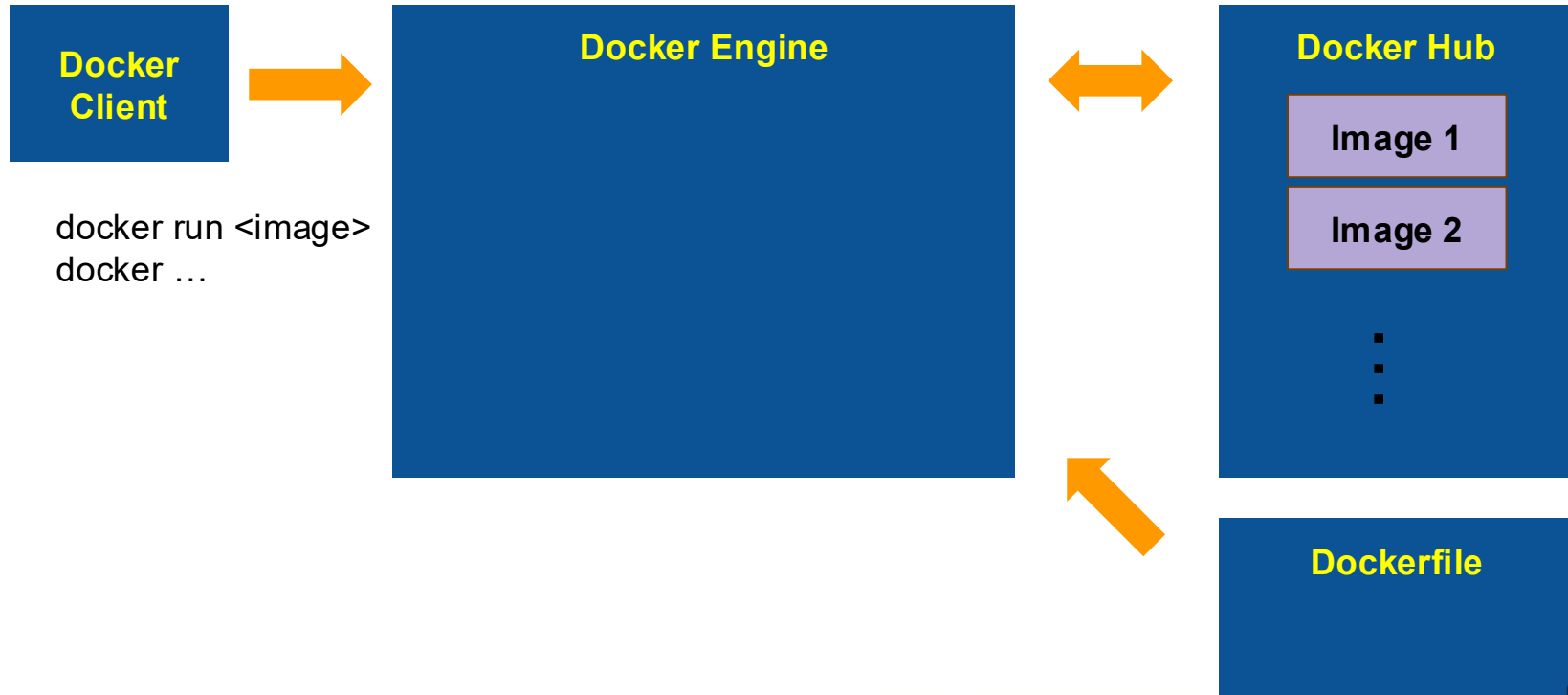
Docker Architecture and Workflow



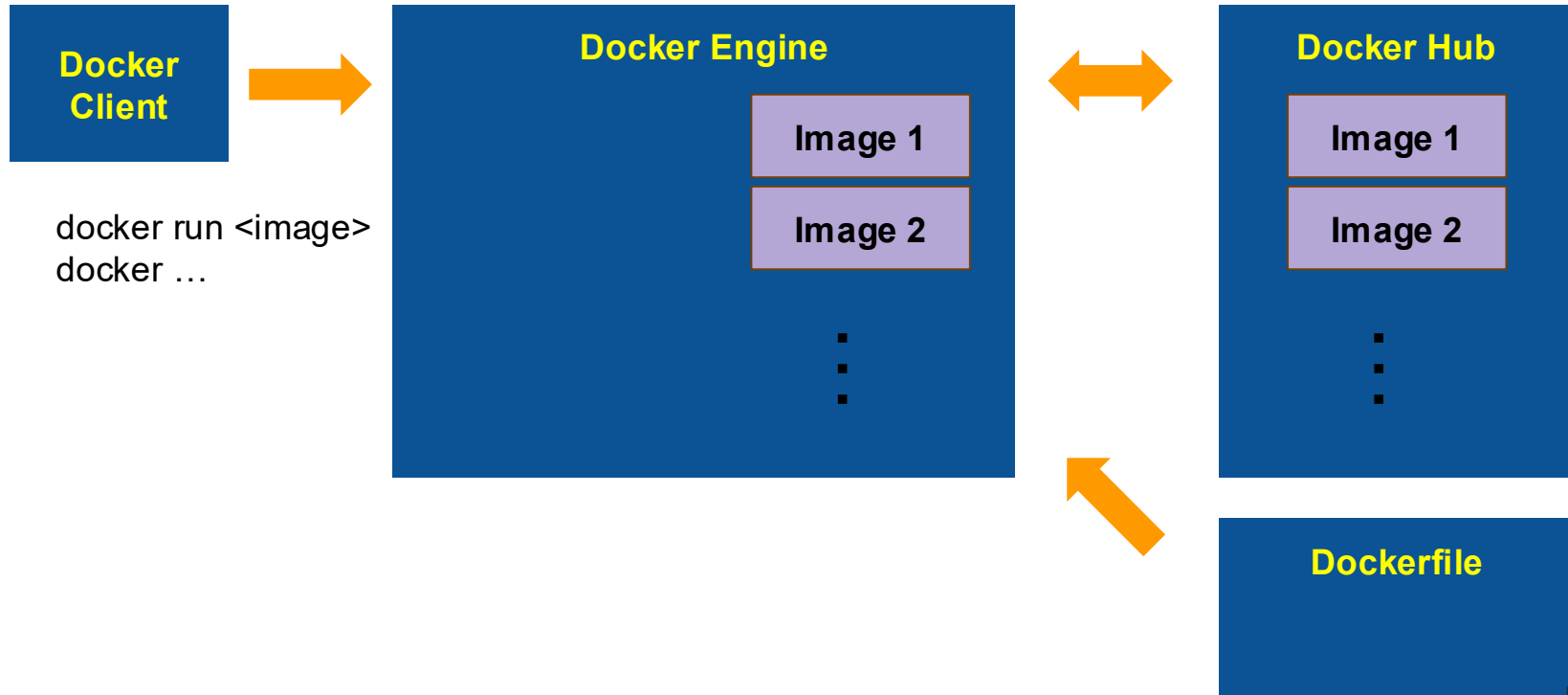
Docker Architecture and Workflow



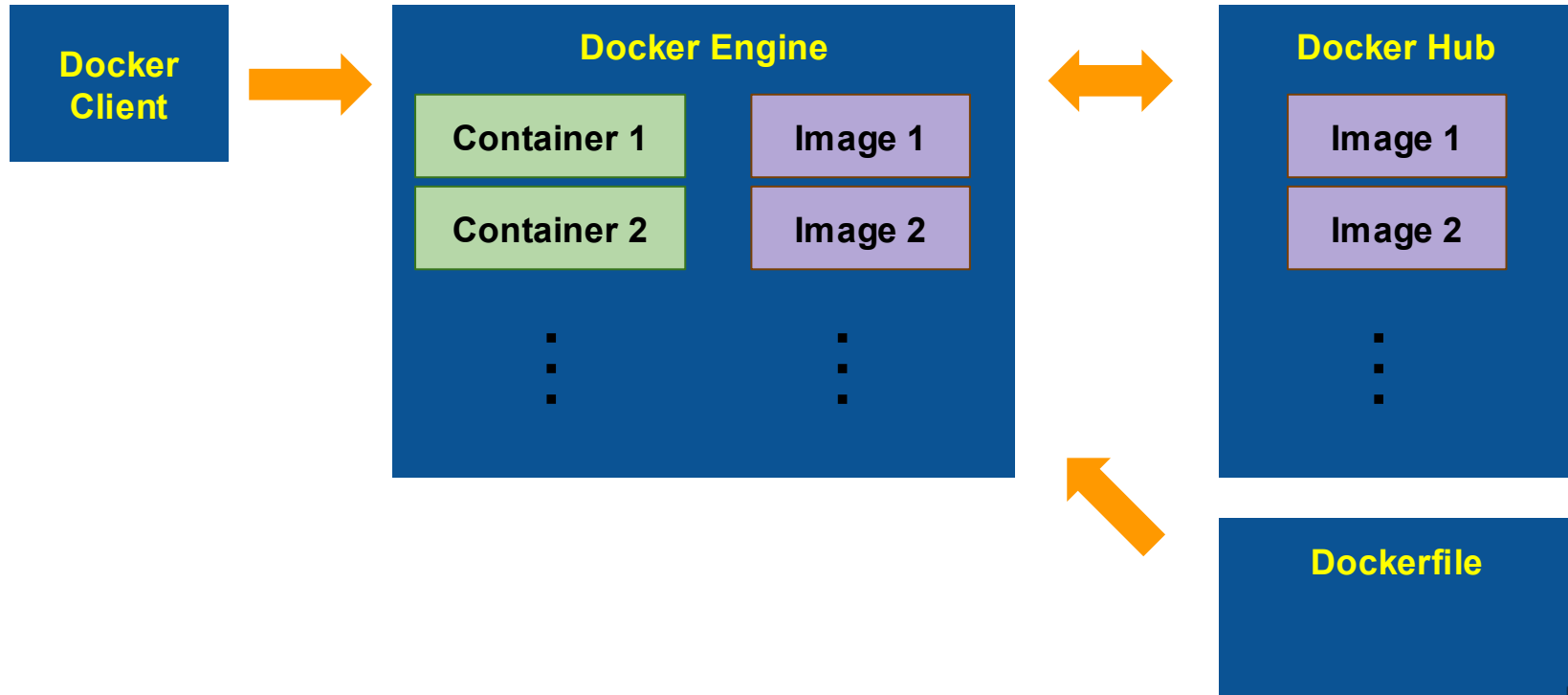
Docker Architecture and Workflow



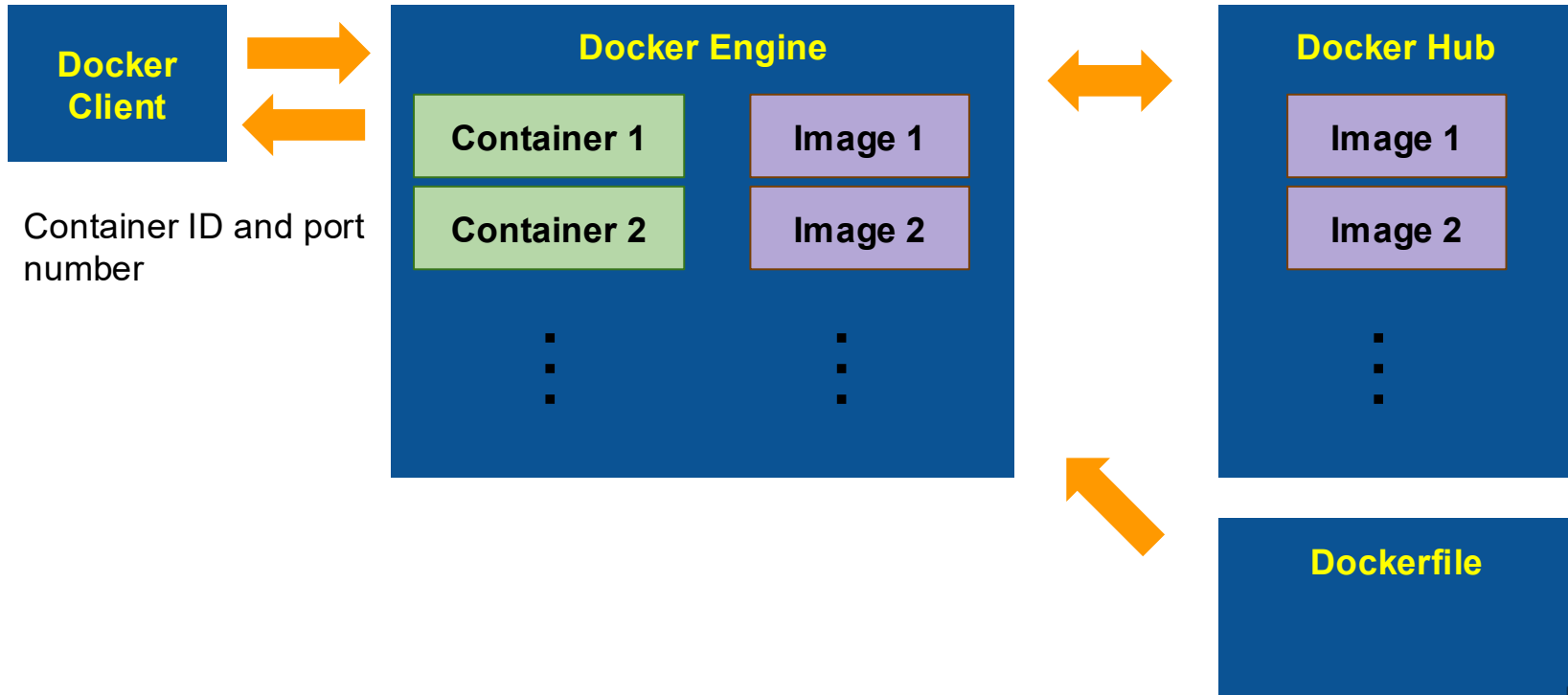
Docker Architecture and Workflow



Docker Architecture and Workflow



Docker Architecture and Workflow



Installing Docker

docker.com



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Develop faster. Run anywhere.

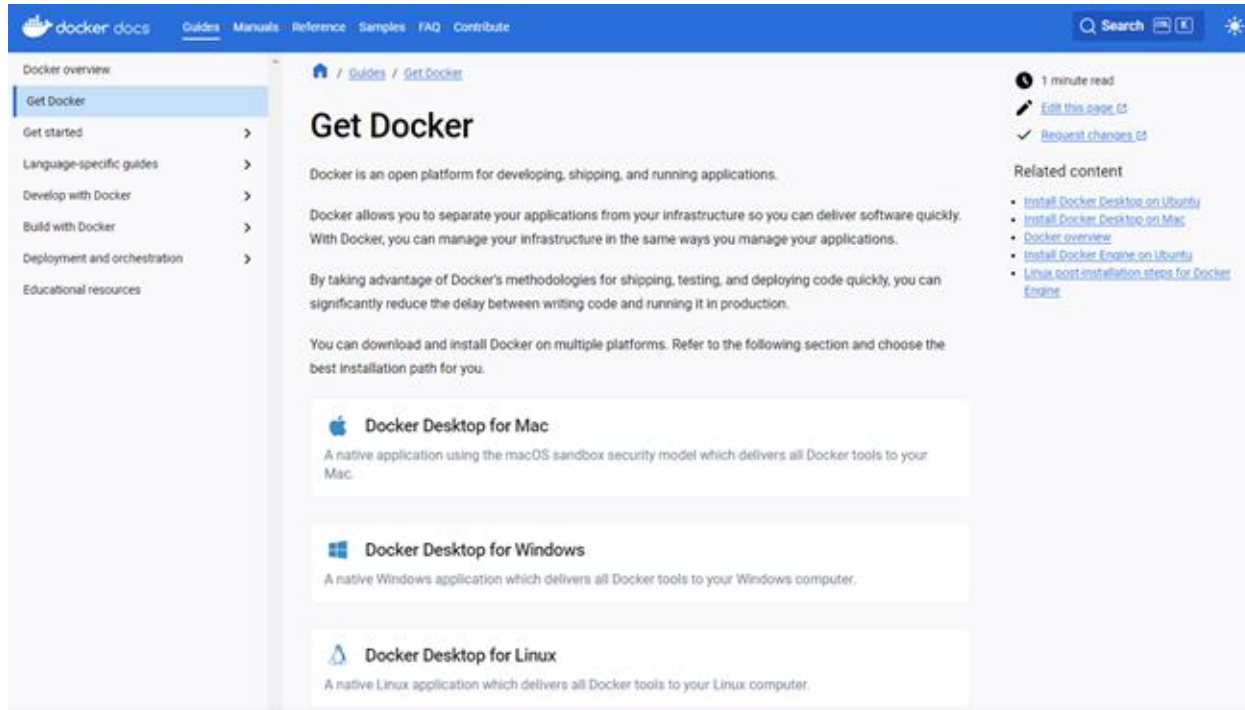
The most-used tool in Stack Overflow's [2023 Developer Survey](#)



Get started

Installing Docker

<https://docs.docker.com/get-docker/>



The screenshot shows the Docker documentation website. The header is blue with the 'docker docs' logo and navigation links: Guides, Manuals, Reference, Samples, FAQ, and Contribute. A search bar is on the right. The left sidebar lists 'Docker overview' and 'Get Docker' (highlighted). The main content area is titled 'Get Docker' and includes a breadcrumb 'Guides / Get Docker'. The text explains that Docker is an open platform for developing, shipping, and running applications, allowing users to separate applications from infrastructure. It mentions that Docker allows managing infrastructure in the same ways as applications. A paragraph states that by taking advantage of Docker's methodologies for shipping, testing, and deploying code quickly, users can significantly reduce the delay between writing code and running it in production. Another paragraph says users can download and install Docker on multiple platforms, referring to the following section for the best installation path. Below this, there are three sections for Docker Desktop: 'Docker Desktop for Mac' (a native application using the macOS sandbox security model), 'Docker Desktop for Windows' (a native Windows application), and 'Docker Desktop for Linux' (a native Linux application). On the right side of the page, there is a '1 minute read' indicator, links to 'Edit this page' and 'Request changes', and a 'Related content' section with links to 'Install Docker Desktop on Ubuntu', 'Install Docker Desktop on Mac', 'Docker overview', 'Install Docker Engine on Ubuntu', and 'Linux post-installation steps for Docker Engine'.

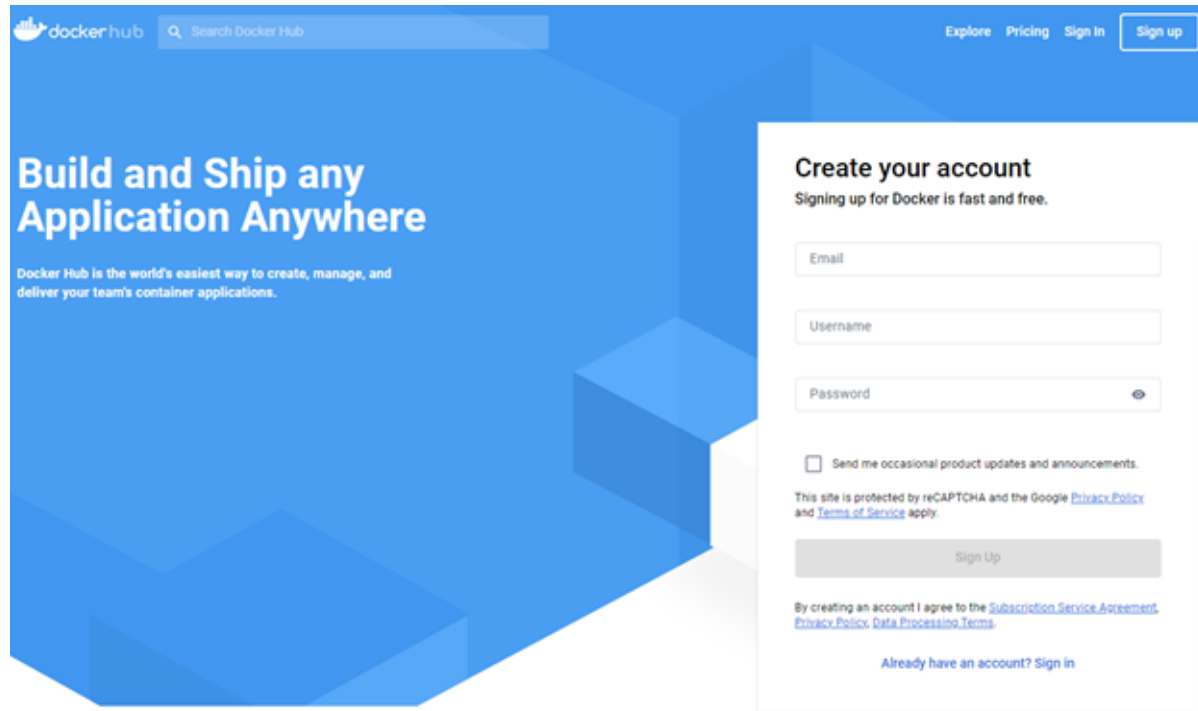
Setup Videos

- [windows](#)
- [macOS](#)



Docker Hub

<https://hub.docker.com/>



The screenshot shows the Docker Hub homepage with a blue background and geometric shapes. On the right, a white 'Create your account' form is overlaid. The form includes fields for Email, Username, and Password, a checkbox for product updates, and a 'Sign Up' button. It also contains links to the Privacy Policy and Terms of Service.

Build and Ship any Application Anywhere

Docker Hub is the world's easiest way to create, manage, and deliver your team's container applications.

Create your account
Signing up for Docker is fast and free.

Email

Username

Password

☐ Send me occasional product updates and announcements.

This site is protected by reCAPTCHA and the Google [Privacy Policy](#) and [Terms of Service](#) apply.

Sign Up

By creating an account I agree to the [Subscription Service Agreement](#), [Privacy Policy](#), [Data Processing Terms](#).

Already have an account? [Sign in](#)

VS Code: Install Docker Extension



Verify your installation

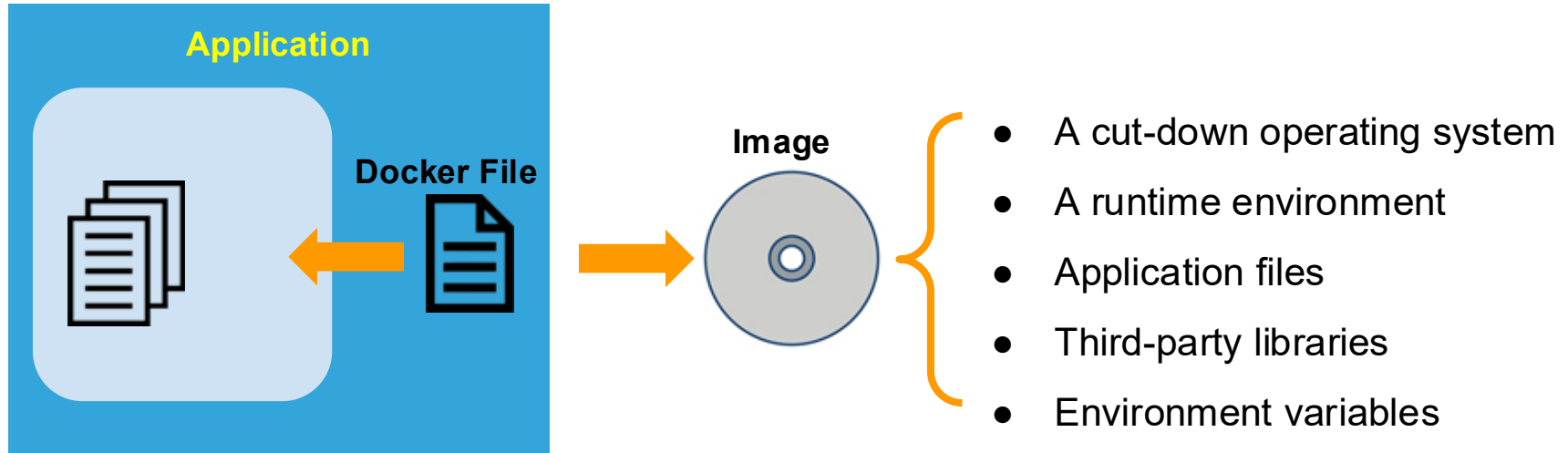
Open a terminal and run:

```
docker version
```

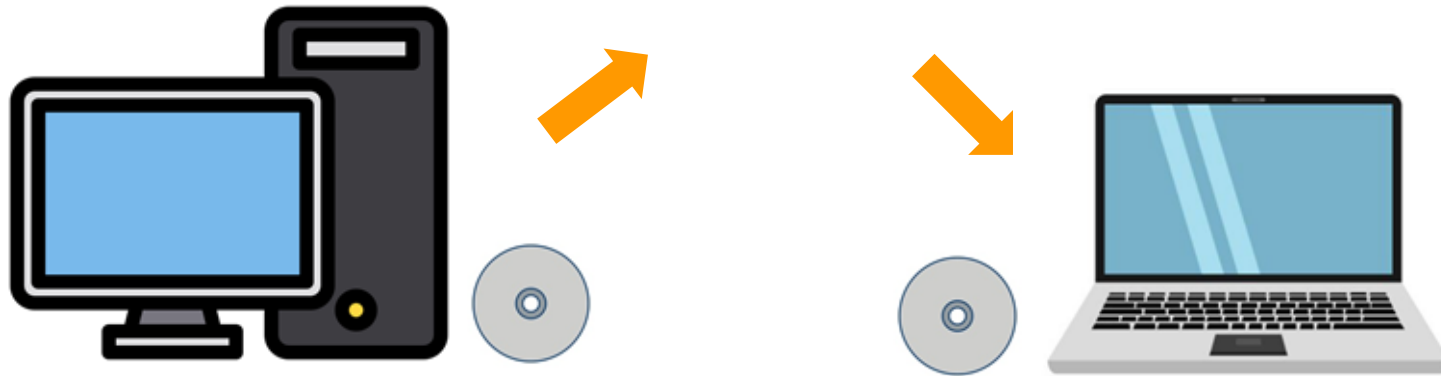
```
C:\Users\jazz1>docker version
Client:
 Cloud integration: v1.0.29
 Version:          20.10.22
 API version:      1.41
 Go version:       go1.18.9
 Git commit:       3a2c30b
 Built:            Thu Dec 15 22:36:18 2022
 OS/Arch:          windows/amd64
 Context:          default
 Experimental:     true

Server: Docker Desktop 4.16.3 (96739)
Engine:
 Version:          20.10.22
 API version:      1.41 (minimum version 1.12)
 Go version:       go1.18.9
 Git commit:       42c8b31
 Built:            Thu Dec 15 22:26:14 2022
 OS/Arch:          linux/amd64
 Experimental:     false
containerd:
 Version:          1.6.14
 GitCommit:       9ba4b250366a5ddde94bb7c9d1def331423aa323
runc:
 Version:          1.1.4
 GitCommit:       v1.1.4-0-g5fd4c4d
docker-init:
 Version:          0.19.0
 GitCommit:       de40ad0
```

Development Workflow

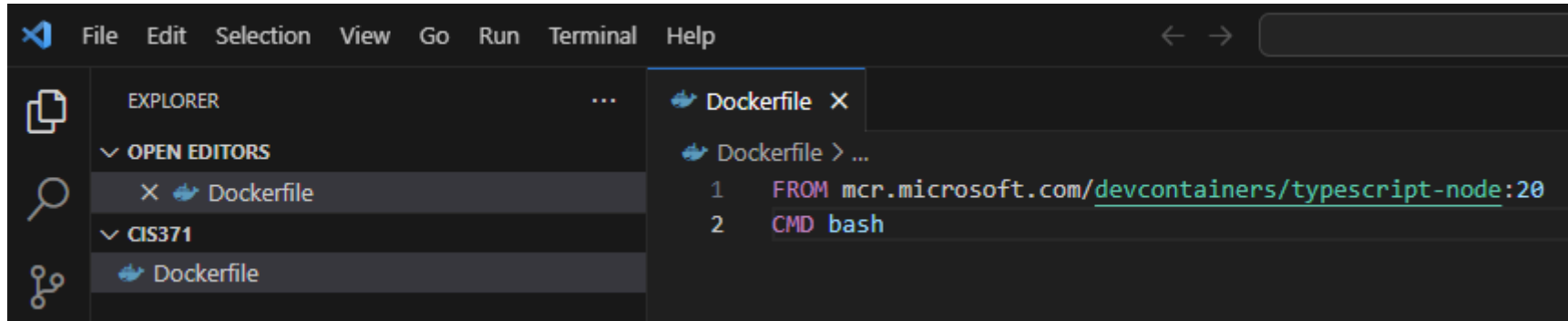


Development Workflow



Example: Build NodeJS + TypeScript Env. Write Dockerfile

Docker image in Docker hub



The screenshot shows the Visual Studio Code interface. The Explorer sidebar on the left shows a file named 'Dockerfile' under the 'CIS371' folder. The main editor window displays the content of the 'Dockerfile' with two lines of code:

```
1 FROM mcr.microsoft.com/devcontainers/typescript-node:20
2 CMD bash
```

Microsoft Devcontainers

Best practices for writing Dockerfiles

Build Image

```
docker build -t your_dockerhub_id/docker_image_name .
```

```
PS C:\Users\jazz1\Documents\workspace\cis371> docker build -t jazz14jazz/cis371 .
[+] Building 47.1s (5/5) FINISHED
=> [internal] load build definition from Dockerfile
=> transferring dockerfile: 956
=> [internal] load .dockerignore
=> transferring context: 28
=> [internal] load metadata for mcr.microsoft.com/devcontainers/base:ubuntu
=> [1/1] FROM mcr.microsoft.com/devcontainers/base:ubuntu@sha256:fd079b61fb4e955b48d2dd025bbae616316994bcd46c193196c6e1664a5d387f
=> resolve mcr.microsoft.com/devcontainers/base:ubuntu@sha256:fd079b61fb4e955b48d2dd025bbae616316994bcd46c193196c6e1664a5d387f
=> sha256:fd079b61fb4e955b48d2dd025bbae616316994bcd46c193196c6e1664a5d387f 1.61kB / 1.61kB
=> sha256:bc84842d3d31ef92e5b48d9957cd0c185d76cc1a8133b8a1620a5e5709f0d164 1.81kB / 1.81kB
=> sha256:46ddc7dca15b0ef0a7c9a2c331fa86f1e1c7221920a72d11c8b8a9c3e90e64 8.85kB / 8.85kB
=> sha256:9d19e268ed77bcf6716e6658e1b08384a71d6f2f8a1a2085619cf7c7b316f 38.13MB / 38.43MB
=> sha256:97968f1f93442a3d45f8304f0cc622b4416879f6c09142fab244ca2083677b 7.12MB / 7.12MB
=> sha256:f8b6dd6570d4a81e537726304ef811f569326941c91a80fc335780e7fd0f72d 411B / 411B
=> sha256:cd0990909289f0b2b0e46bcece4169f9d97fb133bf0226c342c7b0f8227174 135B / 135B
=> sha256:24c81f40e4d021e7470eae557e4153624f0fed939c9f92bfbdd756a50871aca 224B / 224B
=> sha256:9c7fc777f0efb08ea8c382a361d123c403e15c897ec34ef46bc27858ba3a6d84 234B / 234B
=> sha256:2438db792cdaa919ed6d0431a8830d204ed25b3f3fbb974901f6fca9f1fc2c3 74.61MB / 74.61MB
=> sha256:90d3ec78c1338c6e45c8849a3a94a517c56d3143c946132076bdu90e1dd75b 149.75MB / 149.75MB
=> extracting sha256:9d19e268ed77bcf6716e6658e1b08384a71d6f2f8a1a2085619cf7c7b316f 2.1s
=> extracting sha256:97968f1f93442a3d45f8304f0cc622b4416879f6c09142fab244ca2083677b 0.8s
=> extracting sha256:f8b6dd6570d4a81e537726304ef811f569326941c91a80fc335780e7fd0f72d 0.8s
=> extracting sha256:cd0990909289f0b2b0e46bcece4169f9d97fb133bf0226c342c7b0f8227174 0.8s
=> extracting sha256:24c81f40e4d021e7470eae557e4153624f0fed939c9f92bfbdd756a50871aca 0.8s
=> extracting sha256:9c7fc777f0efb08ea8c382a361d123c403e15c897ec34ef46bc27858ba3a6d84 5.1s
=> extracting sha256:2438db792cdaa919ed6d0431a8830d204ed25b3f3fbb974901f6fca9f1fc2c3 5.4s
=> exporting to image
=> exporting layers
=> writing image sha256:b0947b3cf6187e99c3de07c8c02c3087ba55d6578d02e874a770c01777f6485
=> naming to docker.io/jazz14jazz/cis371
PS C:\Users\jazz1\Documents\workspace\cis371> docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
jazz14jazz/cis371	latest	b0947b3cf618	5 weeks ago	661MB

Check Image

```
docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
dockerhub_id/image_name	latest	1b971acc297e	5 weeks ago	1.68GB

Docker Desktop [Update to latest](#) [Ctrl+K](#) [jazz14...](#)

Images [Give feedback](#)

Image is a read-only template with instructions for creating a Docker container. [Learn more](#)

[Local](#) [Hub](#) [Artifactory](#) [EARLY ACCESS](#)

7.34 GB / 12.99 GB in use 6 images Last refresh: about 2 hours ago

☐	Name	Tag	Status	Created	Size	Actions
☐	jazz14jazz/cis371	latest	In use	about 1 month ago	1.67 GB	
	1b971acc297e					

Create a Container(Do This only Once)

```
/* Create a container using docker run */  
docker run --name conatiner_name -it -v  
path to your local workspace:path to workspace in container your_dockerhub_id/docker_image_name  
root → / $
```



**Mount local workspace to
a workspace in the container.**

Create a Container(Do This only Once)

```
/* Create a container using docker run */  
docker run --name cis371 -it -v /c/Users/jazz1/Documents/workspace/cis371:/workspace  
jazz14jazz/cis371  
root → / $
```

Some Docker commands

```
/* Check running containers */
```

```
docker ps
```

```
/* Check all containers */
```

```
docker ps -a
```

```
/* Start container */
```

```
docker start container_name
```

```
/* Stop container */
```

```
docker stop container_name
```

```
/* Re-enter a running container */
```

```
docker exec -it container_name bash
```

Verify the NodeJS + TypeScript Env.

- node: for running JavaScript in a non-browser environment
- npm (Node Package Manager): for installing JS/TS libraries
- npx: Node package runner tool
- tsc: TypeScript Compiler

Verify the NodeJS + TypeScript Env.

```
/* Re-enter a running container */  
docker exec -it cis371 bash  
root → / $
```

```
/* Verify following installations in your container */  
root → / $ node -v  
v20.5.0  
root → / $ npm -v  
9.8.1  
root → / $ npx -v  
9.8.1  
root → / $ tsc -v  
Version 5.1.6
```

Push Docker Image to Dockerhub

```
docker push dockerhub_id/image_name
```

```
Using default tag: latest
```

```
The push refers to repository [docker.io/dockerhub_id/image_name]
```

```
b575e57f29c8: Pushed
```

```
bd96d63ee3b6: Pushed
```

```
f20b22c3fa07: Pushed
```

```
5f70bf18a086: Mounted from jupyter/datascience-notebook
```

```
63c178e39ea1: Pushed
```

```
cde4ef3850e2: Pushed
```

```
b578f477cd5d: Pushed
```

```
b298f9991a11: Pushed
```

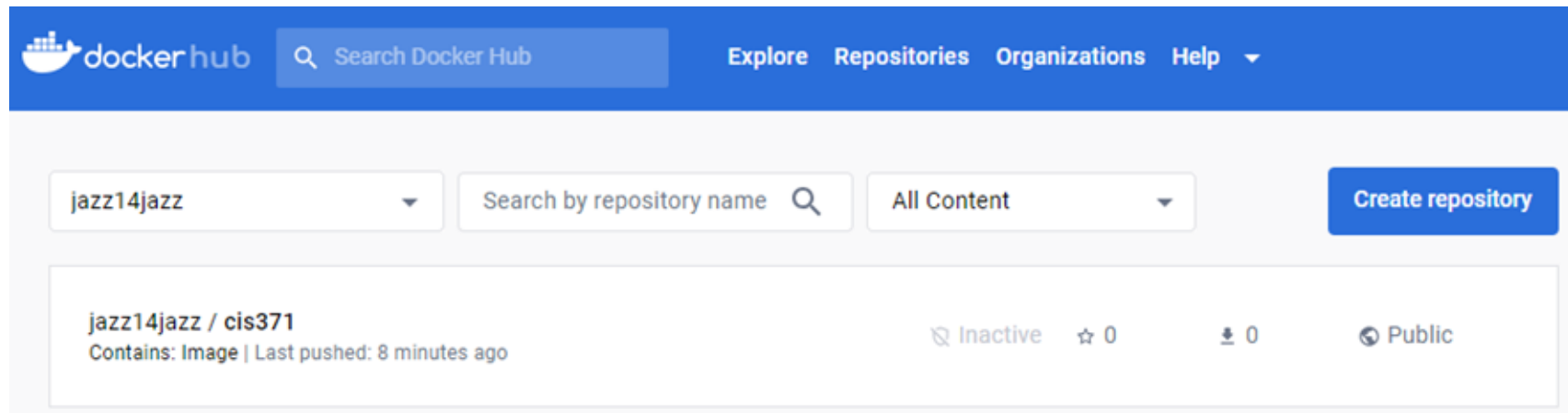
```
c94dc8fa3d89: Pushed
```

```
latest: digest:
```

```
sha256:e71aa9afb60fcb1c7f0b9223e349856dd52ca754502859726c8029422fa5bc3c size: 5141
```

Pull Docker Image

```
docker pull dockerhub_id/image_name
```

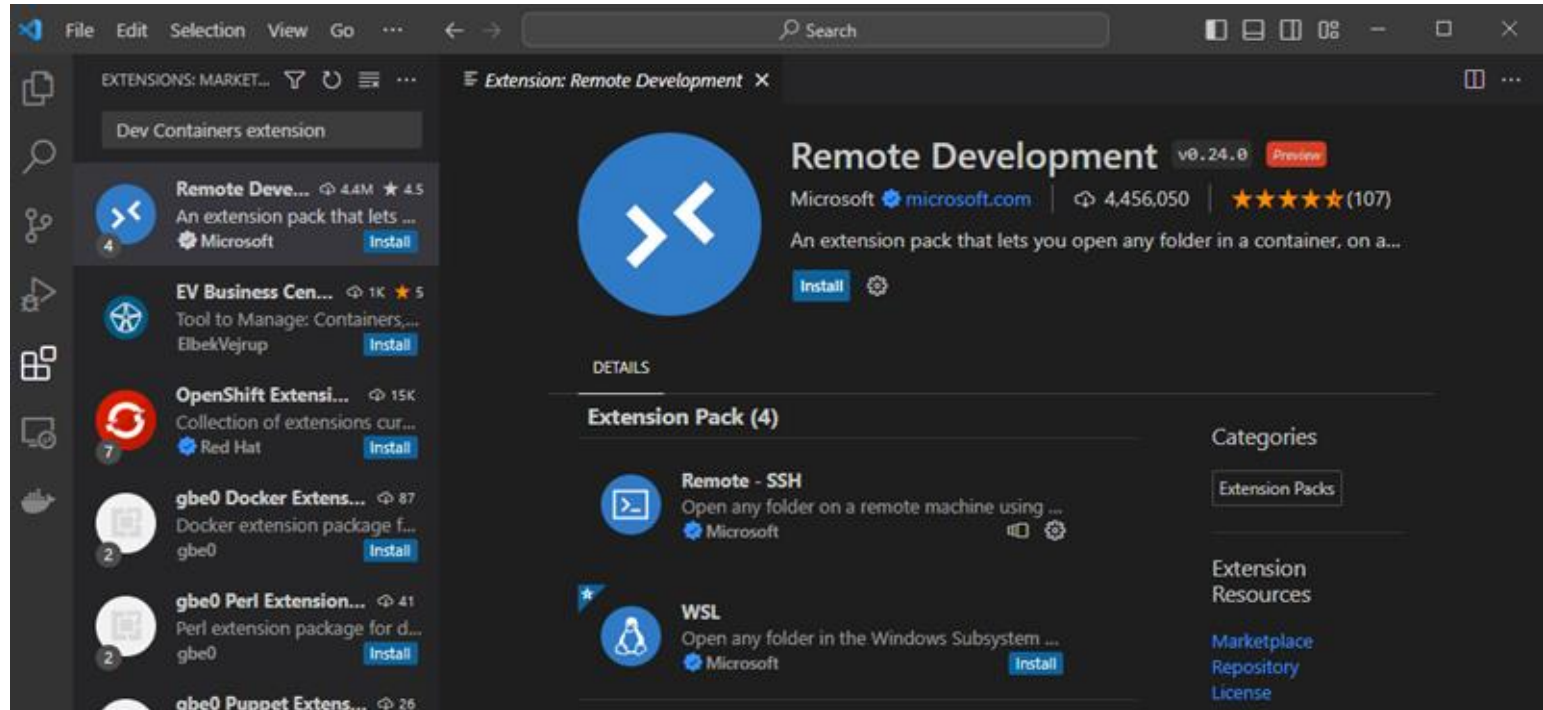


More Docker Commands

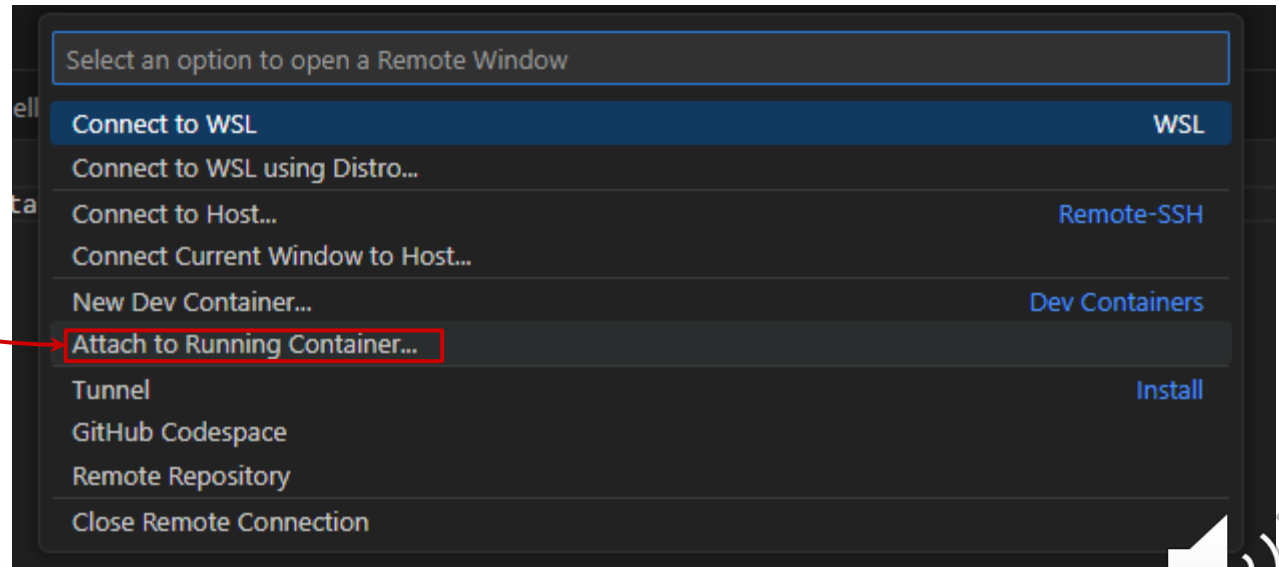
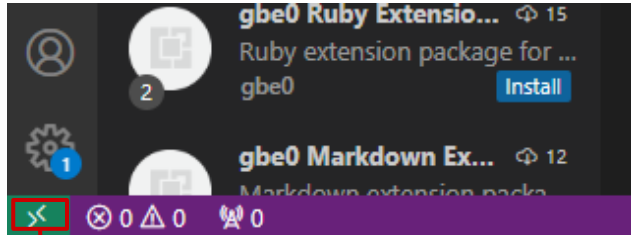
[All Commands](#)

[Cheat Sheet](#)

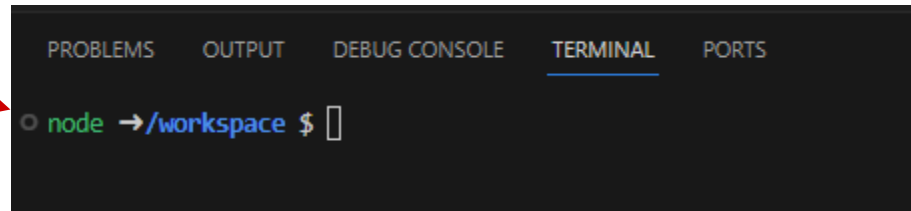
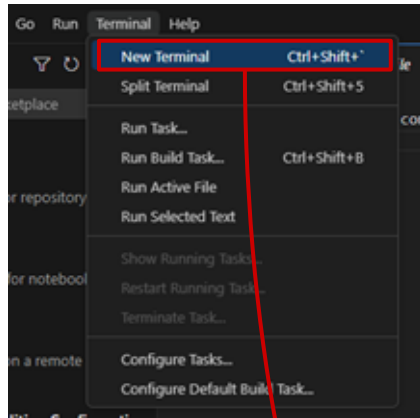
VS Code: Install Remote Development Extension



VS Code: Open the container in VS Code



VS Code: Open a terminal in VS Code



Code Formatting: Prettier

<https://www.youtube.com/watch?v=DqfQ4DPnRqI>

